

DOE Procurement Auditor

What it is, how it works, and what its numbers do and do not support. Everything below is measured, not estimated.

58,523 DOE CONTRACTS	\$53.8B CONTRACT VALUE	0.927 ROLLING AUC	25 PRIORITY CASES
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PART ONE – THE GROUND

What DOE procurement looks like, and what the tool is for.

How do DOE contracts work?

Every NYC Department of Education contract records the **award method** — how DOE chose the vendor. We grade those methods into five tiers of competitive intensity. Across **58,523** contracts worth **\$53.8B** (FY2010–FY2026), the picture is very lopsided.

TIER	CONTRACTS	SHARE	VALUE	WHAT IT MEANS
Informal	44,614	76.2%	\$3.2B	Small purchases and pre-qualified lists — competed lightly
Formal	6,137	10.5%	\$23.2B	Full solicitation: sealed bid, RFP
Renewal	3,994	6.8%	\$7.7B	Option exercised on an existing contract
Discretionary	2,437	4.2%	\$13.9B	No-bid by choice: sole source, negotiated, emergency
Statutory	1,332	2.3%	\$5.8B	No-bid <i>by law</i> — mandates, grants, other governments

The tier that matters is **discretionary**: only **4.2%** of contracts but **\$13.9B** — cases where DOE could have competed the award and chose not to.

What does this tool do?

Two things. It **scores** any contract's characteristics and returns how likely DOE would be to award work like that without a formal solicitation. And it produces a **review queue**: a ranked worklist of real contracts that don't fit the pattern.

It runs entirely in a browser from a static file — no server, no network, no login. Roughly **420 KB** total.

Who uses it, and what is its purpose?

Oversight staff who need to decide *which files to pull*: the Comptroller's office, DOE's own auditors, IBO, Council oversight, investigative reporters. DOE has let **\$13.9B** in discretionary awards over the period — averaging **\$820M** a year, and **\$1.5B** a year since FY2021. Nobody can read all of it.

The purpose is **triage, not judgement**. The tool narrows **2,387** no-bid awards to **25** worth an afternoon — and every one of those **25** still needs a human to read the contract file.

PART TWO — THE MECHANISM

What kind of model, trained on what, flagging what.

What kind of model is it?

A **logistic regression** — **16** standardised features and an intercept. Deliberately the simplest thing that works: it ships as **17** numbers, so an auditor can recompute any score by hand and defend it in a hearing.

We tested a gradient booster on identical features. It scored *worse* — **0.907** mean against the regression's **0.927**, with six times the volatility. Here the transparent model is simply the better one.

What data contributed to this?

Two open records, no API keys, nothing proprietary.

SOURCE	VOLUME	ROLE
Checkbook NYC contracts	58,523	Award method, amount, term, industry, registration dates — the model and the queue
NYC Clerk eLobbyist	79,678	Filings, narrowed to 2,000 naming DOE and 951 about procurement — one corroborating check

The model is fitted on the **9,694** contracts above **\$25,000** where DOE actually chose how much competition to run. The **43,418** small purchases priced at *exactly* **\$25,000** are excluded: the price cap gives the award method away, and a model handed them scores a meaningless **0.982** by learning to recognise the ceiling.

What else are contracts flagged on, and what do those mean?

One primary signal plus four corroborating checks. **1,545** contracts trip at least one; **82** trip two or more. In New York City a contract is **registered** by the Comptroller — the final approval that makes it payable, and it is meant to happen before work starts.

CHECK	CASES	WHAT IT MEANS
Unusual for its industry	25	The primary signal. A no-bid award the model reads as one DOE would normally have put out to bid
Lobbied DOE	32	Vendor filed procurement lobbying against DOE in the award year or the one before — the filing names the officials
Registered after it expired	762	Approved <i>after</i> its own end date: the work was done before the contract formally existed
Registered a year+ late	677	Work began a year or more before approval. The bar is a year because 92.6% are late at all
Long no-bid term	139	A no-bid award running four years or more — how one decision compounds

Only the first drives the ranking. The rest are corroboration: a case tripping two independent checks is a better use of time than one scoring marginally lower on a single model.

PART THREE – THE CREDIBILITY

How good it is, and why this build rather than another.

What is the performance? How accurate is it?

Rolling AUC **0.927 ± 0.012** across **9** independent time periods, range **0.907–0.944**. In plain terms: hand it one competed and one no-bid contract and it ranks them correctly about **93%** of the time.

It is never tested on data it has seen. Each fold trains on fiscal years up to a cut-off, skips the next year entirely, then tests on the two after that. Nine folds rather than one split, because one split hides instability — an earlier version of this model published **0.964** from a single split and swung between **0.685** and **0.961** when re-run properly.

Say this part out loud: **contract term length does most of the work**. Term alone scores **0.901**; strip all term features out and the rest of the model manages **0.823**. Procurement rules tie term length to award method, so part of that accuracy is closer to a definition than a discovery.

What is the selling point? Why is it unique?

Every award is judged against its own industry, never the whole file. The industries are not comparable — Professional Services goes to bid far more readily than Goods — so the flag threshold lands in a completely different place in each:

INDUSTRY	NO-BID RATE	FLAGGED BELOW
Goods	56.6%	33.68%
Human services	42.8%	6.99%
Standardized services	33.9%	13.51%
Professional services	10.3%	0.54%

That is a **62×** spread. Any single cut-off would have flagged whole industries and missed others entirely.

Nothing that happens after the award goes into the model. Registration timing predicts well, and we removed it anyway. Feeding it in taught the model that "no-bid and registered very late" is *normal* — and **0** of those contracts reached the priority list. With it out, **8** of the top **26** are late-registered. We accepted a lower headline number for a tool that finds more.

The rejected ideas are published too. Award timing carries nothing (**0.621** AUC, wildly unstable). Vendor history carries nothing an auditor could look up. M/WBE status carries nothing — removing it *improved* the model. Those negative results are in the repository so nobody re-derives them.

The sentence to lead with, and the one to close on

A flag is a reason to read a file, not a finding. Awarding without competition is lawful and often correct — sometimes one firm genuinely can do the work. Lobbying is lawful, disclosed, and commoner among large vendors precisely *because* they are large. A long term can be the efficient choice.

And the honest limit: the model learns what DOE *does*, not what it *should* do. Where a whole category is routinely sole-sourced, that becomes the expected pattern and will never be flagged. The tool finds departures from DOE's own habits — it cannot tell you whether those habits are correct.